

Elementary Number Theory Exercises

Q&A Notes

Question 1

Prove that if $a > b > 0$ and $c = a + b$, then $c \bmod a = b$.

Answer 1

Since $c = a + b$, write c in division-theorem form with divisor a :

$$c = 1 \cdot a + b.$$

Because $a > b > 0$, we have

$$0 \leq b < a.$$

So b is the unique remainder when c is divided by a . Therefore,

$$\boxed{c \bmod a = b.}$$

Question 2

Prove that there are infinitely many primes.

Answer 2

Assume, for contradiction, that there are only finitely many primes:

$$p_1, p_2, \dots, p_k.$$

Form the integer

$$N = p_1 p_2 \cdots p_k + 1.$$

For each i ,

$$N \bmod p_i = 1,$$

so no p_i divides N . But $N > 1$, so N must have some prime divisor. That prime divisor is not in the list, contradicting the assumption that the list contained all primes.

Therefore,

$$\boxed{\text{there are infinitely many primes.}}$$

Question 3

Prove that if $a \mid b$ and $b \mid c$, then $a \mid c$.

Answer 3

Since $a \mid b$, there is an integer m such that

$$b = am.$$

Since $b \mid c$, there is an integer n such that

$$c = bn.$$

Substitute $b = am$:

$$c = (am)n = a(mn).$$

Because $mn \in \mathbb{Z}$, this means

$$a \mid c.$$

Question 4

Prove that if p is prime and $0 < k < p$, then $\gcd(k, p) = 1$.

Answer 4

Because p is prime, its only positive divisors are

$$1 \quad \text{and} \quad p.$$

Any common divisor of k and p must divide p , so it must be either 1 or p .

But $p \nmid k$, since $0 < k < p$. Therefore the only possible positive common divisor is 1. Hence

$$\gcd(k, p) = 1.$$

Question 5

Prove Corollary 31.5: for all positive integers n, a, b , if $n \mid ab$ and $\gcd(a, n) = 1$, then $n \mid b$.

Answer 5

Since $\gcd(a, n) = 1$, Bézout's theorem gives integers x, y such that

$$ax + ny = 1.$$

Multiply both sides by b :

$$abx + nby = b.$$

Now $n \mid ab$, so $n \mid abx$. Also,

$$nby = n(by),$$

so $n \mid nby$. Therefore n divides the sum:

$$n \mid (abx + nby).$$

Since $abx + nby = b$, we get

$$n \mid b.$$

Question 6

Prove that if p is prime and $0 < k < p$, then $p \mid \binom{p}{k}$. Conclude that for all integers a, b and all primes p ,

$$(a + b)^p \equiv a^p + b^p \pmod{p}.$$

Answer 6

First,

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}.$$

Equivalently,

$$k!(p-k)! \binom{p}{k} = p!.$$

The right-hand side contains a factor p , so

$$p \mid k!(p-k)! \binom{p}{k}.$$

For $0 < k < p$, none of the integers in $k!$ or $(p-k)!$ is divisible by p . Since p is prime, Euclid's lemma forces

$$p \mid \binom{p}{k}.$$

Now use the binomial theorem:

$$(a+b)^p = \sum_{k=0}^p \binom{p}{k} a^{p-k} b^k.$$

The first and last terms are

$$a^p \quad \text{and} \quad b^p.$$

Every middle coefficient $\binom{p}{k}$, where $1 \leq k \leq p-1$, is divisible by p . Hence all middle terms vanish modulo p . Therefore,

$$(a+b)^p \equiv a^p + b^p \pmod{p}.$$

Question 7

Prove that if a and b are positive integers such that $a \mid b$, then

$$(x \bmod b) \bmod a = x \bmod a$$

for any x . Under the same assumptions, prove that

$$x \equiv y \pmod{b} \implies x \equiv y \pmod{a}.$$

Answer 7

Let

$$x = qb + r, \quad 0 \leq r < b.$$

Then

$$x \bmod b = r.$$

Since $a \mid b$, write

$$b = at$$

for some integer t . Then

$$x = qat + r.$$

So x and r differ by a multiple of a :

$$x - r = qat.$$

Therefore,

$$x \equiv r \pmod{a}.$$

Thus x and $x \bmod b$ have the same remainder modulo a , so

$$(x \bmod b) \bmod a = x \bmod a.$$

For the second part, if

$$x \equiv y \pmod{b},$$

then

$$b \mid (x - y).$$

Since $a \mid b$, transitivity of divisibility gives

$$a \mid (x - y).$$

Therefore,

$$x \equiv y \pmod{a}.$$

Question 8

For any integer $k > 0$, an integer n is a k th power if there exists an integer a such that $a^k = n$. Furthermore, $n > 1$ is a nontrivial power if it is a k th power for some integer $k > 1$. Show how to determine whether a given β -bit integer n is a nontrivial power in time polynomial in β .

Answer 8

It is enough to test exponents

$$2 \leq k \leq \beta,$$

because if $n = a^k$ and $a \geq 2$, then

$$2^k \leq a^k = n < 2^\beta,$$

so $k < \beta$.

For each $k = 2, 3, \dots, \beta$, use binary search to find whether there is an integer a such that

$$a^k = n.$$

The possible range for a is

$$2 \leq a \leq 2^{\lceil \beta/k \rceil}.$$

During binary search, compute a^k using repeated squaring and compare it with n .

There are at most β exponents to test, each binary search has polynomially many bit operations, and exponentiation/comparison on β -bit integers is polynomial in β . Therefore the whole procedure runs in polynomial time.

Test all $2 \leq k \leq \beta$ using integer k th-root search.

Question 9

Prove the following basic gcd identities:

$$\gcd(a, b) = \gcd(b, a),$$

$$\gcd(a, b) = \gcd(-a, b),$$

$$\gcd(a, b) = \gcd(|a|, |b|),$$

$$\gcd(a, 0) = |a|,$$

$$\gcd(a, ka) = |a| \quad \text{for any } k \in \mathbb{Z}.$$

Answer 9

The first identity holds because common divisors are symmetric:

$$d \mid a \text{ and } d \mid b \iff d \mid b \text{ and } d \mid a.$$

Thus

$$\boxed{\gcd(a, b) = \gcd(b, a).}$$

The second holds because

$$d \mid a \iff d \mid (-a).$$

So a and $-a$ have the same divisors:

$$\boxed{\gcd(a, b) = \gcd(-a, b).}$$

The third follows by applying the sign rule to both arguments:

$$\boxed{\gcd(a, b) = \gcd(|a|, |b|).}$$

For the fourth, every positive divisor of a divides 0, since

$$0 = 0 \cdot d.$$

The largest positive divisor of a is $|a|$, so

$$\boxed{\gcd(a, 0) = |a|.$$

For the fifth, $|a| \mid a$ and $|a| \mid ka$, so $|a|$ is a common divisor. No common divisor of a and ka can be larger than $|a|$, because it must divide a . Therefore,

$$\boxed{\gcd(a, ka) = |a|.$$

Question 10

Show that the gcd operator is associative. That is, prove that for all integers a, b, c ,

$$\gcd(a, \gcd(b, c)) = \gcd(\gcd(a, b), c).$$

Answer 10

Let

$$g_1 = \gcd(a, \gcd(b, c)).$$

Then $g_1 \mid a$ and $g_1 \mid \gcd(b, c)$. Since $g_1 \mid \gcd(b, c)$, we have

$$g_1 \mid b \quad \text{and} \quad g_1 \mid c.$$

So g_1 is a common divisor of a, b, c .

Conversely, any common divisor of a, b, c divides b and c , so it divides $\gcd(b, c)$. If it also divides a , then it divides $\gcd(a, \gcd(b, c))$. Therefore g_1 is exactly the greatest common divisor of all three numbers.

Similarly,

$$g_2 = \gcd(\gcd(a, b), c)$$

is also the greatest common divisor of a, b, c . Hence

$$\gcd(a, \gcd(b, c)) = \gcd(\gcd(a, b), c).$$

Question 11

Prove Theorem 31.8: unique factorization.

Answer 11

Assume $n > 1$ has two prime factorizations:

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s.$$

Since $p_1 \mid n$, we have

$$p_1 \mid q_1 q_2 \cdots q_s.$$

By Euclid's lemma, p_1 divides some q_j . Since both are prime, this forces

$$p_1 = q_j.$$

Move q_j to the front of the right-hand product and cancel it with p_1 . Then repeat the same argument for $p_2, p_3, \dots$

Every prime factor on the left must match a prime factor on the right. Neither side can have extra unmatched primes, because then a product of primes greater than 1 would have to equal 1.

Thus the two factorizations contain exactly the same primes, counted with repetition. Grouping repeated primes gives the same prime powers:

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r} \text{ uniquely.}$$

Question 12

Give efficient algorithms for dividing a β -bit integer by a shorter integer and for taking the remainder of a β -bit integer when divided by a shorter integer. The algorithms should run in time $\Theta(\beta^2)$.

Answer 12

Use binary long division.

Let N be the β -bit dividend and let d be the shorter positive divisor. Scan the bits of N from most significant to least significant. Maintain a partial remainder r .

```
q = 0
r = 0
for each bit bit_i of N from left to right:
    r = 2*r + bit_i
    q = 2*q
    if r >= d:
        r = r - d
        q = q + 1
return q, r
```

At each step, the algorithm shifts r , compares with d , and possibly subtracts d . These opera-

tions cost $O(\beta)$ bit operations. There are β bits, so the total running time is

$$\boxed{\Theta(\beta^2)}.$$

The quotient is q , and the remainder is $r = N \bmod d$.

Question 13

Give an efficient algorithm to convert a given β -bit binary integer to a decimal representation. Argue that if multiplication or division of integers whose length is at most β takes time $M(\beta)$, then binary-to-decimal conversion takes time $\Theta(M(\beta) \lg \beta)$.

Answer 13

Use divide and conquer.

Split the binary integer N into two halves:

$$N = H \cdot 2^m + L,$$

where L contains the lower m bits and H contains the remaining upper bits.

Recursively convert H and L to decimal. Then combine:

$$N = H \cdot 2^m + L.$$

The multiplication by 2^m and the addition are performed on integers of length at most β , costing $O(M(\beta))$ at the top level.

The recurrence is

$$T(\beta) = 2T(\beta/2) + O(M(\beta)).$$

Across each recursion level, the total multiplication/addition cost is $O(M(\beta))$, and there are $O(\lg \beta)$ levels. Therefore,

$$\boxed{T(\beta) = \Theta(M(\beta) \lg \beta)}.$$